Reflective Journal

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One of the biggest breakthroughs for our team was a better understanding of how supervised machine learning actually works. Before this lab, machine learning seemed like a black box where you enter data and somehow get predictions. After working with the wine dataset, we started to view supervised learning as a process of learning from an answer key. The model uses known outcomes, such as wine cultivars, to learn patterns between chemical features and classifications. Looking at the heatmap also helped us realize the model's predictions are based on relationships in the data, such as alcohol content and color intensity, rather than random or magical patterns.

One challenge our team faced was learning the scikit-learn functions and understanding what each line of code was doing. At first, the number of unfamiliar functions felt overwhelming. To work through this, we stopped focusing on every line individually and instead connected each section of code to a six-step machine workflow. For example, once we understood that the train-test split represented the data-splitting stage, the entire script became easier to follow. This experience showed us that understanding the overall process is just as important as understanding the overall process as it is just as important as understanding the code itself.

The lab also raised new questions for our group. We were surprised that Logistic Regression achieved a higher accuracy score (88.9%) than the Decision Tree model (83.3%). We initially assumed that a more complex model would automatically perform better. Instead, we learned that model performance depends on the data set and the problem being solved. This made us wonder how data scientists determine which algorithm is most appropriate when starting a new project.

Our thinking about the model evaluation also changed. We now understand that high performance on training data does not necessarily mean a model is successful because it may simply be memorizing the data. The train-test split became one of the most important concepts from this lab because it allows us to test the model on unseen data and determine whether it can generalize to real-world situations.